

University of Jeddah

College of Computer Science and Engineering

Department of Software Engineering

CCSW321 – WEB DEVELOPMENT

Assignment #2: Web Services, APIs, AJAX, and DOM

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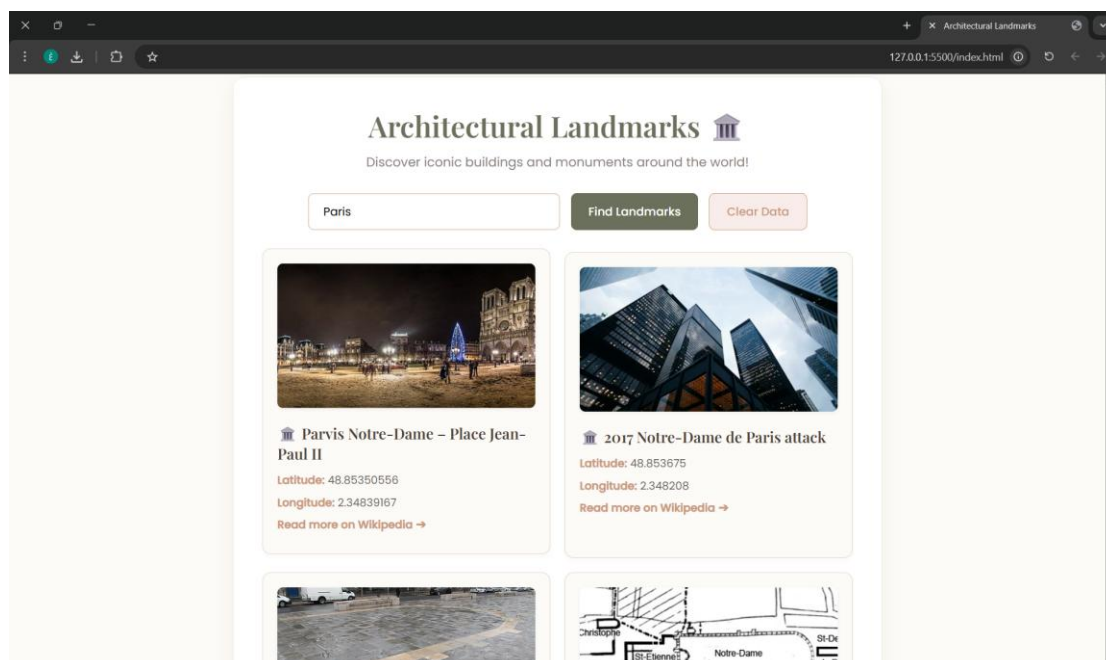
Project Description:

- **Chosen API:** I selected a combination of **OpenStreetMap API** (for Geocoding) and **Wikipedia API** (for fetching landmarks). This choice is unique as it focuses on Architectural Heritage.
- **Event Types Used:**
 1. click: On the Search and Clear buttons.
 2. keyup: To allow searching by pressing the "Enter" key.
 3. input: To clear error messages dynamically while typing.
- **Data Points Displayed:** For each landmark, the application displays: (Image, Title, Latitude, Longitude, and a direct Wikipedia link).

Screenshots:

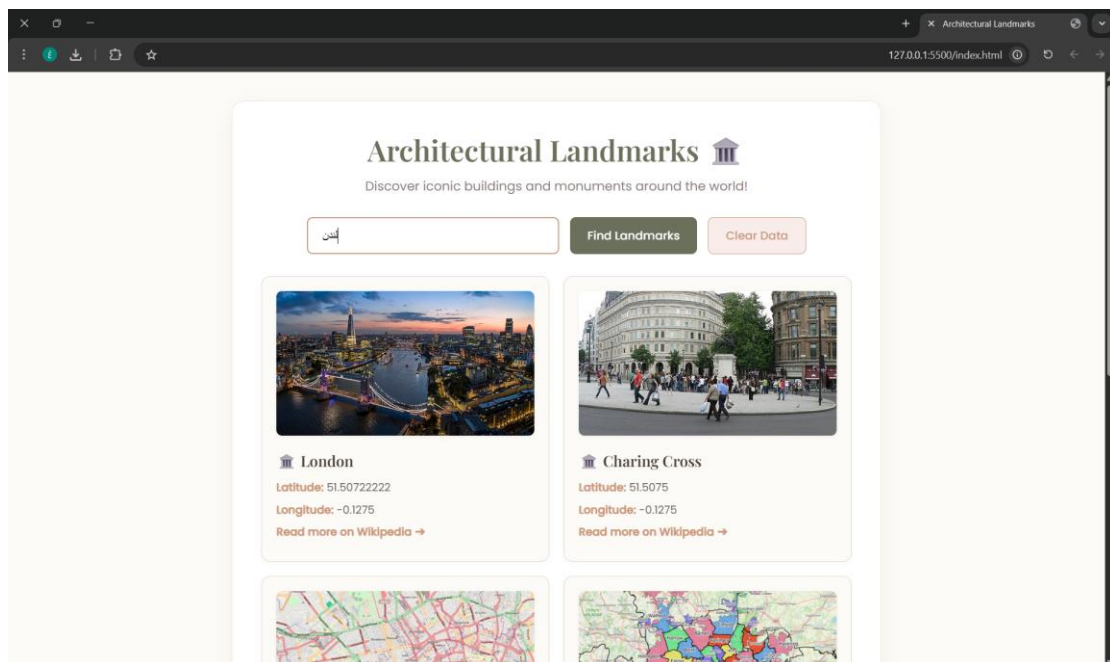
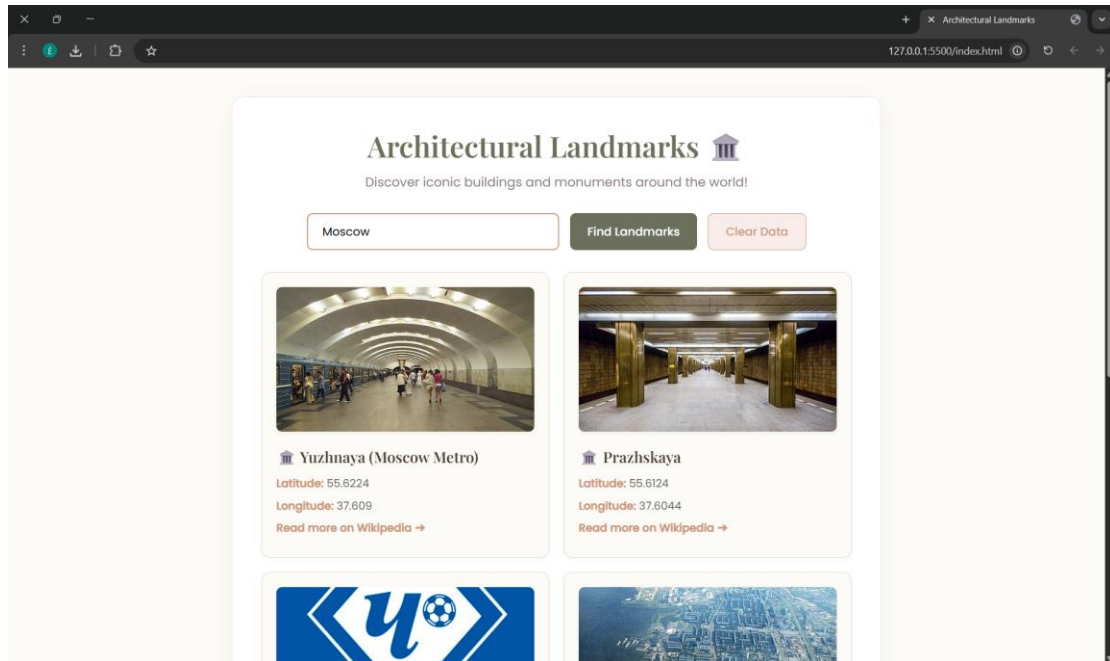
Successful Search:

Displaying architectural landmarks with images and coordinates after a successful AJAX fetch.



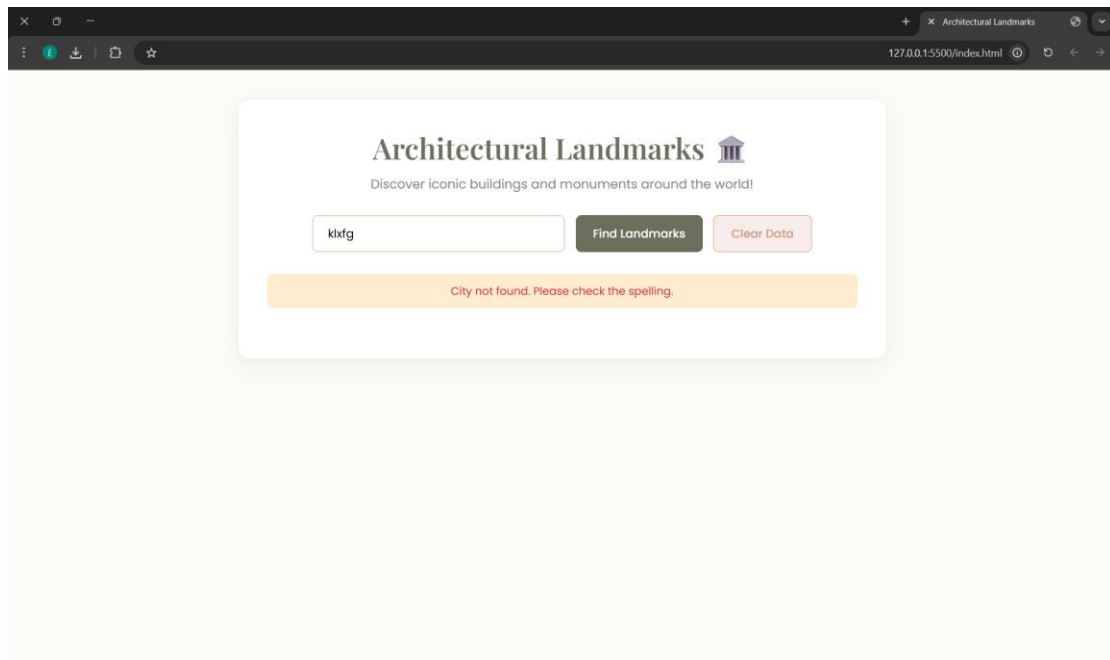
Multilingual Support:

The application supports both Arabic and English queries, retrieving accurate geographical data for both.



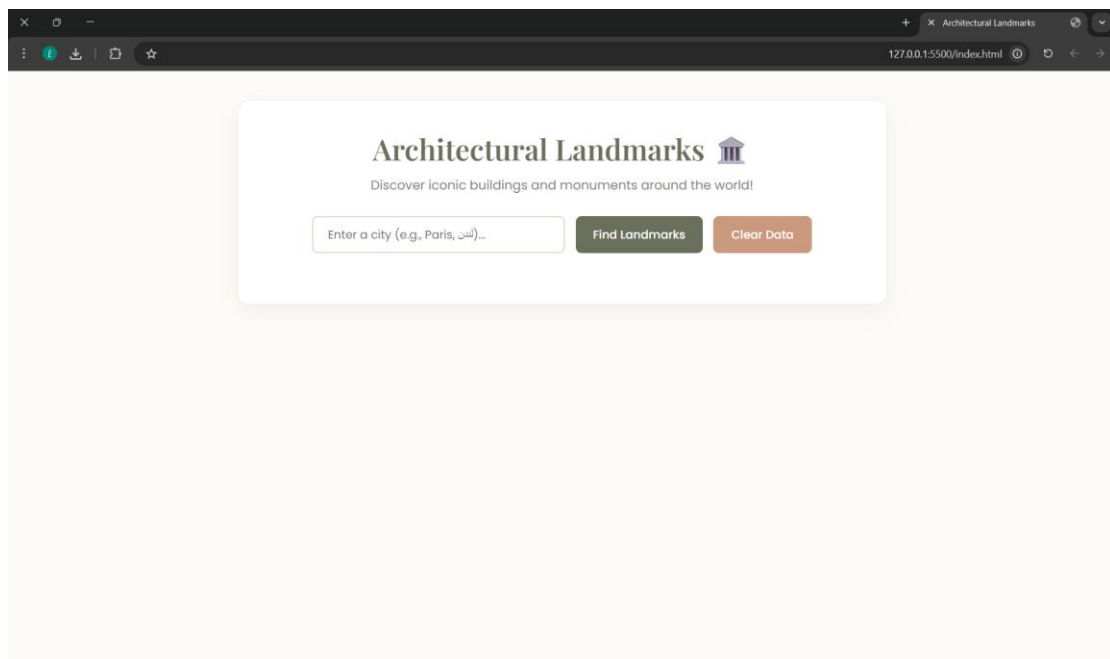
Error Handling

Proper error handling when the user provides empty or invalid input



Clear Data

Demonstrating dynamic DOM manipulation by removing all generated elements upon clicking the Clear button.



Code:

```
index.html - Assignment 2 ghaada2317174 - Visual Studio Code

1 <!DOCTYPE html>
2 <html lang="en">
3 <head>
4   <meta charset="UTF-8">
5   <meta name="viewport" content="width=device-width, initial-scale=1.0">
6   <title>Architectural Landmarks</title>
7   <link rel="stylesheet" href="style.css">
8 </head>
9 <body>
10
11   <div class="container">
12     <h1>Architectural Landmarks</h1>
13     <p>Discover iconic buildings and monuments around the world</p>
14
15     <div class="form-group">
16       <input type="text" id="cityInput" placeholder="Enter a city (e.g., Paris, لندن)..." autocomplete="off">
17       <button id="searchBtn" class="btn">Search</button>
18     </div>
19
20     <div id="errorMessage" class="error-hidden"></div>
21
22     <div id="landmarksContainer">
23     </div>
24   </div>
25
26   <script src="script.js"></script>
27 </body>
28 </html>
```

```
script.js - Assignment 2 ghaada2317174 - Visual Studio Code

1 const cityInput = document.getElementById('cityInput');
2 const searchBtn = document.getElementById('searchBtn');
3 const clearBtn = document.getElementById('clearBtn');
4 const landmarksContainer = document.getElementById('landmarksContainer');
5 const errorMessage = document.getElementById('errorMessage');
6
7 searchBtn.addEventListener('click', () => {
8   const city = cityInput.value.trim();
9   if (city) {
10     fetchLandmarks(city);
11   } else {
12     showError("Please enter a city name first!");
13   }
14 });
15
16 cityInput.addEventListener('keyup', (event) => {
17   if (event.key === 'Enter') {
18     searchBtn.click();
19   }
20 });
21
22 clearBtn.addEventListener('click', () => {
23   landmarksContainer.innerHTML = '';
24   cityInput.value = '';
25   hideError();
26 });
27
28 cityInput.addEventListener('input', hideError);
29
30 async function fetchLandmarks(city) {
31   hideError();
32   landmarksContainer.innerHTML = `<p style="text-align:center; color:#888">Searching for landmarks and images...</p>`;
33
34   try {
35     const geoId = `https://nominatim.openstreetmap.org/search?q=${city}&format=jsonlimit=1`;
36     const geoResponse = await fetch(geoId);
37     const geoData = await geoResponse.json();
38
39     if (geoData.length === 0) {
40       throw new Error("City not found. Please check the spelling.");
41     }
42
43     const lat = geoData[0].lat;
44     const lon = geoData[0].lon;
45
46     const wikiId = `https://en.wikipedia.org/w/api.php?action=query&generator=geosearch&gscoord=${lat}|${lon}&gsradius=10000&gslimit=50&prop=pageimages|coordinates|info`;
47     const wikiResponse = await fetch(wikiId);
48     const wikiData = await wikiResponse.json();
49
50     const pages = wikiData.query?.pages;
```

```
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script.js - Assignment 2 ghada2317174 - Visual Studio Code

# script.js
30 fetchLandmarks
31 async function fetchLandmarks(city) {
32   const pages = wikiData.query?.pages;
33   if (!pages) {
34     throw new Error("No architectural landmarks found nearby.");
35   }
36   landmarksContainer.innerHTML = "";
37   Object.values(pages).forEach(place => {
38     const card = document.createElement('div');
39     card.className = 'landmark-card';
40     const imgUrl = place.thumbnail ? place.thumbnail.source : 'https://images.unsplash.com/photo-1486406146925-c627a92ad1ab?y=88&w=400&auto=format&fit=crop';
41     const placeLat = place.coordinates ? place.coordinates[0].lat : 'N/A';
42     const placeLon = place.coordinates ? place.coordinates[0].lon : 'N/A';
43     const wikiLink = place.fullurl;
44     card.innerHTML = `
45     <img src=${imgUrl} alt=${place.title} class=landmark-img>
46     <h3>${place.title}</h3>
47     <p><strong>Latitude:</strong> ${placeLat}</p>
48     <p><strong>Longitude:</strong> ${placeLon}</p>
49     <p><a href=${wikiLink} target=_blank style=color: #c999e; text-decoration: none; font-weight: bold;>Read more on Wikipedia </a></p>
50     `;
51     landmarksContainer.appendChild(card);
52   });
53   catch (error) {
54     landmarksContainer.innerHTML = "";
55     showError(error.message || "An error occurred while fetching data.");
56   }
57 }
58 function showError(msg) {
59   errorMessage.textContent = msg;
60   errorMessage.className = 'error-visible';
61 }
62 function hideError() {
63   errorMessage.className = 'error-hidden';
64 }
65
```

```
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style.css - Assignment 2 ghada2317174 - Visual Studio Code

# style.css
1 @import url('https://fonts.googleapis.com/css2?family=PlayfairDisplay:ital,weight@400,600,700&family=Poppins:weight@300,400,500&display=swap');
2
3 {
4   box-sizing: border-box;
5   margin: 0;
6   padding: 0;
7 }
8
9 body {
10   font-family: 'Poppins', sans-serif;
11   background-color: #f0f0f0;
12   color: #8a4036;
13   display: flex;
14   justify-content: center;
15   align-items: flex-start;
16   min-height: 100vh;
17   padding: 40px 20px;
18 }
19
20 .container {
21   background-color: #ffffff;
22   max-width: 900px;
23   width: 100%;
24   border-radius: 16px;
25   padding: 40px;
26   box-shadow: 0 10px 30px #8a4036;
27   text-align: center;
28   border: 1px solid #f0eae1;
29 }
30
31 h1 {
32   font-family: 'Playfair Display', serif;
33   color: #8a4036;
34   margin-bottom: 10px;
35   font-size: 2.5rem;
36 }
37
38 p {
39   margin-bottom: 30px;
40   font-size: 1.1rem;
41   color: #8a4036;
42 }
43
44 .form-group {
45   display: flex;
46   flex-wrap: wrap;
47   gap: 15px;
48   justify-content: center;
49   margin-bottom: 30px;
50 }
51
```

```
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style.css - Assignment 2 ghada2317174 - Visual Studio Code

# style.css x scripts index.html
# style.css > %container
51
52 input[type="text"] {
53   padding: 12px 20px;
54   border: 2px solid #c3d5dc;
55   border-radius: 8px;
56   font-size: 1rem;
57   outline: none;
58   width: 100%;
59   max-width: 350px;
60   font-family: 'Poppins', sans-serif;
61   transition: border-color 0.3s;
62 }
63
64 input[type="text"]:focus {
65   border-color: #c39999;
66 }
67
68 button {
69   padding: 12px 24px;
70   border: none;
71   border-radius: 8px;
72   font-size: 1rem;
73   cursor: pointer;
74   font-family: 'Poppins', sans-serif;
75   font-weight: 500;
76   transition: all 0.3s ease;
77 }
78
79 #searchBtn {
80   background-color: #f0f0f0;
81   color: #333;
82 }
83
84 #searchBtn:hover {
85   background-color: #e0e0e0;
86 }
87
88 #clearBtn {
89   background-color: #f0f0f0;
90   color: #c39999;
91   border: 1px solid #c39999;
92 }
93
94 #clearBtn:hover {
95   background-color: #e0e0e0;
96   color: #333;
97 }
98
99 .error-hidden {
100   display: none;
101 }
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style.css - Assignment 2 ghada2317174 - Visual Studio Code

# style.css x scripts index.html
# style.css > %searchBtn
103
104 .error-visible {
105   padding: 12px;
106   border-radius: 8px;
107   margin-bottom: 20px;
108   font-size: 0.95rem;
109 }
110
111
112
113 #landmarkContainer {
114   display: grid;
115   grid-template-columns: repeat(auto-fit, minmax(280px, 1fr));
116   gap: 20px;
117   margin-top: 30px;
118 }
119
120 .landmark-card {
121   background-color: #f0f0f0;
122   border: 1px solid #c3d5dc;
123   border-radius: 12px;
124   padding: 20px;
125   text-align: left;
126   box-shadow: 0 4px 6px #c3d5dc;
127   transition: transform 0.3s ease;
128 }
129
130 .landmark-card:hover {
131   transform: translateY(-5px);
132 }
133
134 .landmark-card h3 {
135   font-family: 'Playfair Display', serif;
136   color: #c3d5dc;
137   margin-bottom: 12px;
138   font-size: 1.3rem;
139 }
140
141 .landmark-card p {
142   font-size: 0.95rem;
143   margin-bottom: 8px;
144   color: #c3d5dc;
145   line-height: 1.5;
146 }
147
148 .landmark-card strong {
149   color: #c3d5dc;
150 }
151
152 .landmark-ing {
153   width: 100%;
154   height: 200px;
155   object-fit: cover;
156   border-radius: 8px;
157   margin-bottom: 15px;
158   box-shadow: 0 4px 6px #c3d5dc;
159 }
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